



Languages

Italian (mother tongue)

English (fluent)

Spanish (beginner)

Core skills

Robotics & control: **KUKA LBR Med**;
Software: **Python, C/C++, Java, MATLAB, Arduino, SolidWorks.**

Lab skills: **sensor** calibration
soldering (incl. SMD), **3D printing**
(FDM), general **mechanical skills.**

Licence of professional engineer.

Scientific writing.

Knee biomechanics.

Driving licence: A, B.

Hobbies & Other activities

Volunteering:

- **Ambulance driver** and **first responder** (2014 - present);
- **BLSD and safety trainer** for internal and external staff;
- Event **organizer** and **cook** (with HACCP) for **country fair (Fa'Festa 2011 - present).**

Personal projects:

- **DIY sous-vide cooker**;
- **Programmable PSU**
- **Full restoration** of a **Piaggio Si.**

Interests:

- **Trekking and mountain biking**;
- **Cooking.**

Lorenzo Maggi

Biomedical Engineer

Hands-on biomedical engineer (PhD) with experience in **surgical robotics, knee biomechanics, and ex-vivo experimentation.** During my PhD I developed a knee kinematic simulator and contributed to the design of **Robin®**, a robotic system for total knee arthroplasty. Strong **maker** mindset (**mechanics, electronics, 3D printing**) and a focus on **translational** research.

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🌐 <https://lorenzo-maggi.github.io/>

Experience & Education

R&D engineer

September 2026 - ongoing

Clover Orthopedics Srl

R&D Robotics and Biomedical engineer

November 2025 - August 2026

Orthokey Italia Srl

- **Co-development** of **Robin®** robotic system for **TKA**;
- **ISO 13485, IEC 60601, IEC 80601** oriented design;
- **KUKA LBR Med** software development;
- Collaboration with **National Research Council** for Murine emotional analysis via **neural networks.**

PhD in Industrial Engineering (cum Laude)

November 2022 - October 2025

Orthokey Italia Srl - University of Florence

- **Development** of an innovative **knee laxity** assessment robotic tool based on **KUKA LBR Med**;
- **Development** of a **force-feedback control** for rehabilitation **hand exoskeleton**;
- **2 years** of Industrial experience at **Orthokey Italia Srl**;
- A brief **summary** of my **PhD**: <https://youtu.be/5DqW-fMwWBU>

Visiting PhD researcher

September 2024 - March 2025

Université Libre de Bruxelles (Prof. B. Innocenti)

- **6-month abroad experience**;
- **Development** of a portable **knee kinematic** simulator for kinematic analysis of different **TKA implants**;
- Implementation of a **PID controller** to simulate **quadriceps tendon loading**;
- Participation in several **cadaveric knee tests** with the kinematic simulator.

MSc in Biomedical engineering (110/110 cum Laude)

October 2022

BSc in Mechanical engineering

February 2021

University of Florence

I hereby authorize the processing of my personal data in accordance with Legislative Decree No. 196 of June 30, 2003 and Article 13 of the EU General Data Protection Regulation (GDPR).